WORKED SOLUTION

STEP I 2013, Question 3

Pure • Vector Algebra

STEP I 2013, Question 3

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/13-s1-q3

The key idea

Everything here is linear, so the cleanest move every time is to write each point out as a combination of $\mathbf{x}$, $\mathbf{y}$ and $\mathbf{z}$ and then subtract. Two points are distinct exactly when the difference of their position vectors is non-zero, and that difference always turns out to be a simple factor times a vector like $\mathbf{x}-\mathbf{y}$. Geometrically, $X*Y$ is a point on the line XY, and when $0<\lambda<1$ it lies on the segment between them.

Solution

Throughout, two points are the same exactly when their position vectors are equal.

Part (i)

By definition $Y*X$ has position vector $\lambda\mathbf{y}+(1-\lambda)\mathbf{x}$, so

$$\begin{aligned}(\lambda\mathbf{x}+(1-\lambda)\mathbf{y})-(\lambda\mathbf{y}+(1-\lambda)\mathbf{x}) &= (2\lambda-1)\mathbf{x}-(2\lambda-1)\mathbf{y}\\ &= (2\lambda-1)(\mathbf{x}-\mathbf{y}).\end{aligned}$$

Since X and Y are distinct, $\mathbf{x}-\mathbf{y}\neq\mathbf{0}$, so this difference is zero if and only if $2\lambda-1=0$. Hence $X*Y$ and $Y*X$ are distinct unless $\lambda=\frac{1}{2}$, when both are the midpoint of XY.

Part (ii)

Expanding each side,

$$\begin{aligned}(X*Y)*Z:&\quad \lambda(\lambda\mathbf{x}+(1-\lambda)\mathbf{y})+(1-\lambda)\mathbf{z}=\lambda^2\mathbf{x}+\lambda(1-\lambda)\mathbf{y}+(1-\lambda)\mathbf{z},\\ X*(Y*Z):&\quad \lambda\mathbf{x}+(1-\lambda)(\lambda\mathbf{y}+(1-\lambda)\mathbf{z})=\lambda\mathbf{x}+\lambda(1-\lambda)\mathbf{y}+(1-\lambda)^2\mathbf{z}.\end{aligned}$$

The $\mathbf{y}$ terms agree, and subtracting,

$$\begin{aligned}\text{difference} &= (\lambda^2-\lambda)\mathbf{x} + ((1-\lambda)-(1-\lambda)^2)\mathbf{z}\\ &= -\lambda(1-\lambda)\mathbf{x} + \lambda(1-\lambda)\mathbf{z}\\ &= \lambda(1-\lambda)(\mathbf{z}-\mathbf{x}).\end{aligned}$$

This is non-zero exactly when each factor is non-zero. So $(X*Y)*Z$ and $X*(Y*Z)$ are distinct if and only if

$$\lambda \neq 0, \qquad \lambda \neq 1 \qquad \text{and} \qquad Z \neq X.$$

Notice that Y plays no part. The excluded values make sense: with $\lambda = 1$ the operation always returns the first point, and with $\lambda = 0$ it always returns the second, so both bracketings give X or Z respectively.

Part (iii)

Write $\mu = 1-\lambda$ to keep things tidy. The right-hand side is

$$\begin{aligned}(X*Z)*(Y*Z): \quad &\lambda(\lambda\mathbf{x}+\mu\mathbf{z}) + \mu(\lambda\mathbf{y}+\mu\mathbf{z})\\ &= \lambda^2\mathbf{x} + \lambda\mu\,\mathbf{y} + \mu(\lambda+\mu)\mathbf{z}\\ &= \lambda^2\mathbf{x} + \lambda\mu\,\mathbf{y} + \mu\,\mathbf{z},\end{aligned}$$

since $\lambda+\mu = 1$. This is exactly the expansion of $(X*Y)*Z$ found in part (ii), so

$$(X*Y)*Z = (X*Z)*(Y*Z)$$

for all points X, Y, Z, as required.

? **Ask yourself** What should the "corresponding result" for $X*(Y*Z)$ look like?

In part (iii) the outer $*Z$ is distributed over the bracket on the left. The mirror image is to distribute $X*$ over the bracket on the right, which suggests trying $(X*Y)*(X*Z)$. Indeed,

$$\begin{aligned}(X*Y)*(X*Z): \quad &\lambda(\lambda\mathbf{x}+\mu\mathbf{y}) + \mu(\lambda\mathbf{x}+\mu\mathbf{z})\\ &= \lambda(\lambda+\mu)\mathbf{x} + \lambda\mu\,\mathbf{y} + \mu^2\mathbf{z}\\ &= \lambda\mathbf{x} + \lambda\mu\,\mathbf{y} + \mu^2\mathbf{z},\end{aligned}$$

which is the expansion of $X*(Y*Z)$ from part (ii). So, for any points X, Y, Z,

$$X*(Y*Z) = (X*Y)*(X*Z).$$

Part (iv)

We claim that P_n has position vector

$$\mathbf{p}_n = \lambda^n\mathbf{x} + (1-\lambda^n)\mathbf{y}.$$

For $n = 1$ this is the definition of $X * Y$. If it holds for $n - 1$, then

$$\begin{aligned}\mathbf{p}_n = \lambda\mathbf{p}_{n-1} + (1-\lambda)\mathbf{y} &= \lambda^n\mathbf{x} + (\lambda - \lambda^n + 1 - \lambda)\mathbf{y} \\ &= \lambda^n\mathbf{x} + (1-\lambda^n)\mathbf{y},\end{aligned}$$

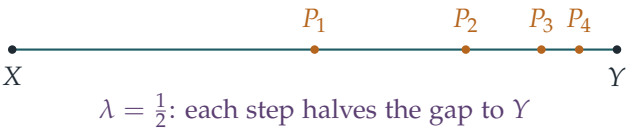
so by induction the formula holds for all $n \geq 1$.

Now measure along the segment:

$$\mathbf{p}_n - \mathbf{x} = (1-\lambda^n)(\mathbf{y} - \mathbf{x}), \qquad \mathbf{y} - \mathbf{p}_n = \lambda^n(\mathbf{y} - \mathbf{x}).$$

Both are positive multiples of the same non-zero vector $\mathbf{y} - \mathbf{x}$, because $0 < \lambda < 1$ gives $0 < \lambda^n < 1$. So P_n lies strictly between X and Y, and

$$XP_n : P_nY = (1-\lambda^n) : \lambda^n.$$



Each step multiplies the remaining distance to Y by λ, so the points march towards Y, and $P_n \to Y$ as $n \to \infty$.

Checks

- In part (i), take $\mathbf{x} = \mathbf{0}$, $\mathbf{y} = \mathbf{1}$ along a line and $\lambda = \frac{1}{3}$: $X * Y$ is at $\frac{2}{3}$ and $Y * X$ at $\frac{1}{3}$, which differ by $\frac{1}{3} = |2\lambda - 1|$.
- In part (ii), with $Z = X$ both bracketings should agree for any λ; our difference $\lambda(1-\lambda)(\mathbf{z} - \mathbf{x})$ vanishes then, as it should.
- In part (iv) with $\lambda = \frac{1}{2}$ and $n = 3$ the ratio is $\frac{7}{8} : \frac{1}{8} = 7 : 1$, and indeed P_3 is at $\frac{7}{8}$ of the way from X to Y (halving the gap three times). With $n = 1$ the ratio $(1-\lambda) : \lambda$ is just the definition of $X * Y$.

Common slips

- Forgetting $Z \neq X$ in part (ii): the answer is not just a condition on λ.
- Giving the ratio the wrong way round in part (iv). Since $\mathbf{p}_n$ carries weight λ^n on $\mathbf{x}$, it is *close* to X when λ^n is large, so the shorter piece XP_n goes with $1 - \lambda^n$.

Beyond STEP

The point $\lambda\mathbf{x} + (1-\lambda)\mathbf{y}$ is a convex (or affine) combination, the building block of convex sets, barycentric coordinates and linear interpolation in computer graphics. Part

(ii) says $*$ is not associative, while part (iii) says it is self-distributive; operations with this property are studied in algebra under names such as racks, quandles and medial algebras. For the linear-algebra viewpoint, see Strang, *Introduction to Linear Algebra*.

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